Week 7-Binary response

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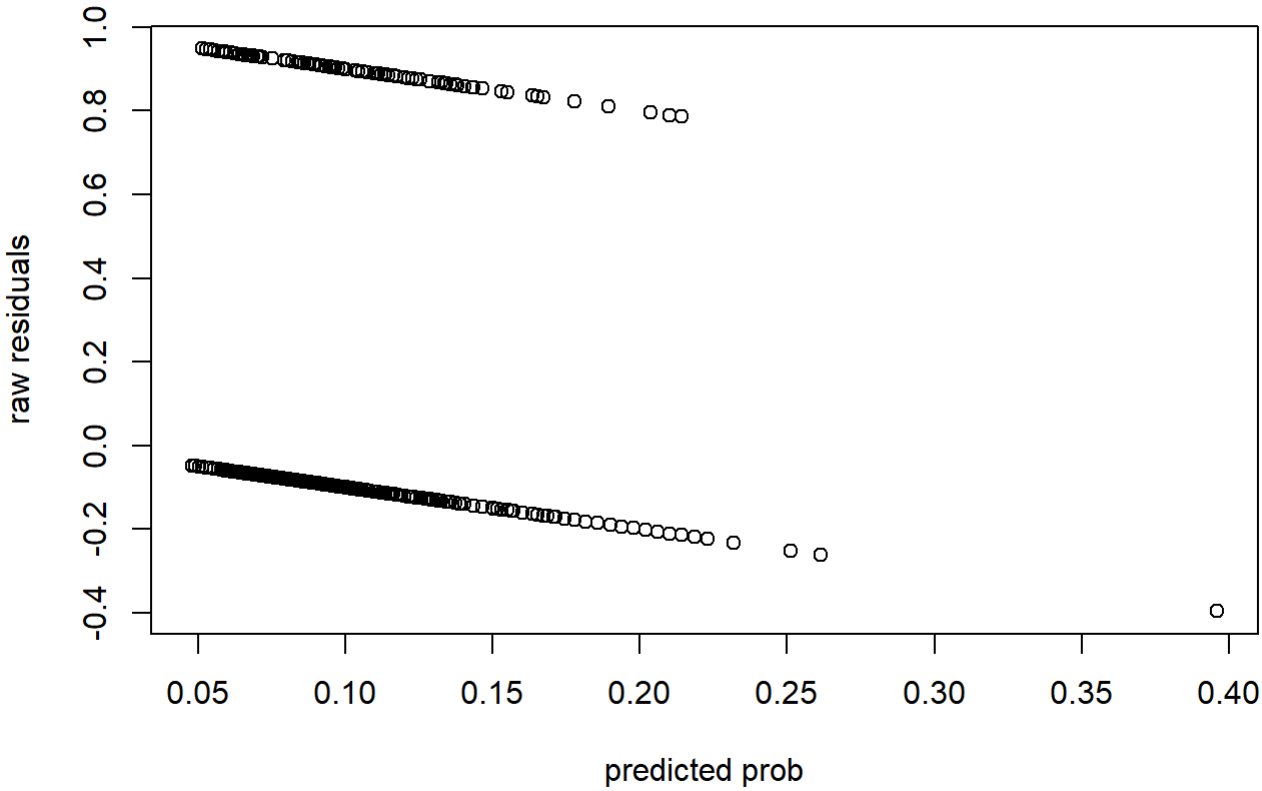
1 Diagnostics (section 2.4)

1.1 Raw residuals

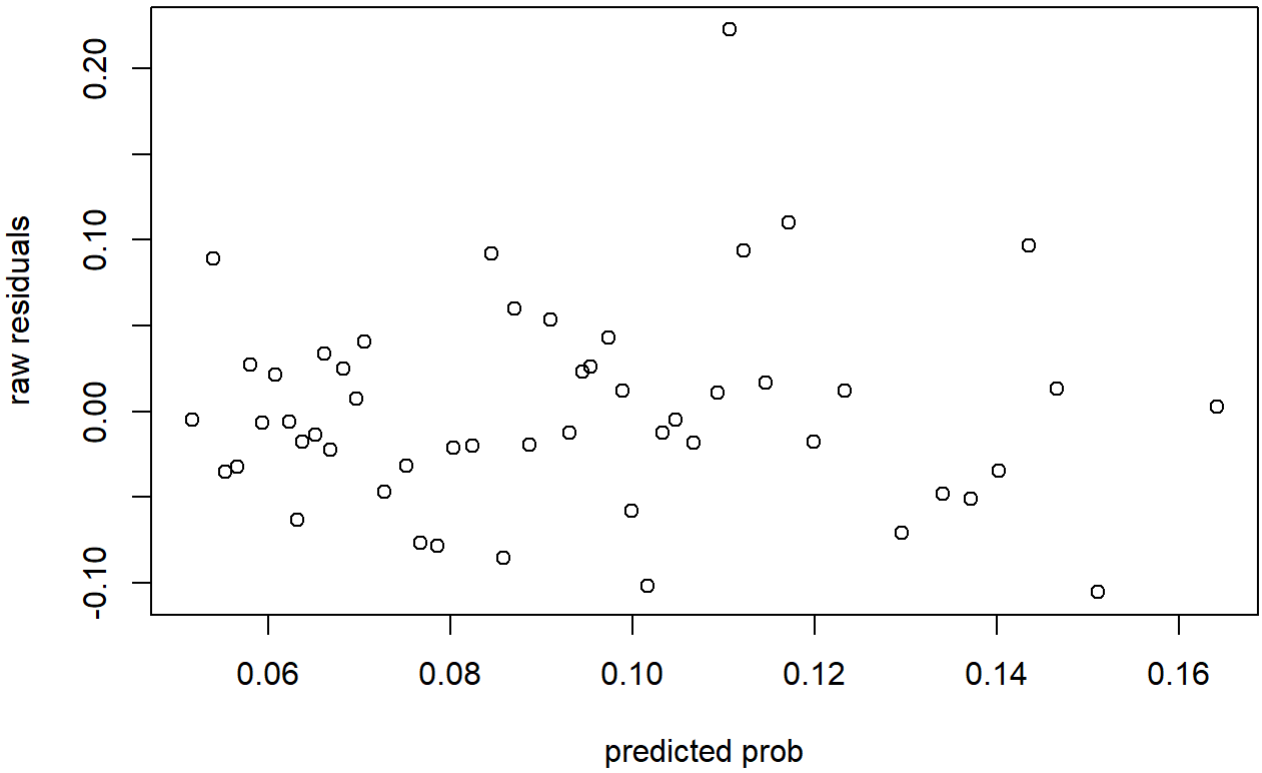
Let $\hat{\eta}_i = x_i^T \hat{\beta}$be the linear predictor for the individual i. The corresponding predicted probability is $\hat{p_i} = \frac{e^{\hat{\eta}_i}}{1+e^{\hat{\eta}_i}}$

$$\text{Raw residuals} : y_i - \hat{p}_i$$

```
data(wcgs, package="faraway")
wcgs$y <- ifelse(wcgs$chd == "no",0,1)
lmod <- glm(chd ~ height + cigs, family = binomial, wcgs)

linpred <- predict(lmod)
predprob <- predict(lmod, type="response")

rawres <- wcgs$y - predprob
plot(predprob, rawres, xlab="predicted prob", ylab="raw residuals")
```



```
# Grouping
quantile.p<-quantile(predprob, (1:100)/101)
group<-cut(predprob, breaks=unique(c(min(predprob)-.1,quantile.p)))

# Calculating Bin Statistics
predprob.bin<-tapply(predprob, group ,mean)
rawres.bin<-tapply(rawres, group, mean)

plot(predprob.bin,rawres.bin,xlab="predicted prob",ylab="raw residuals")
```



1.2 Deviance residuals

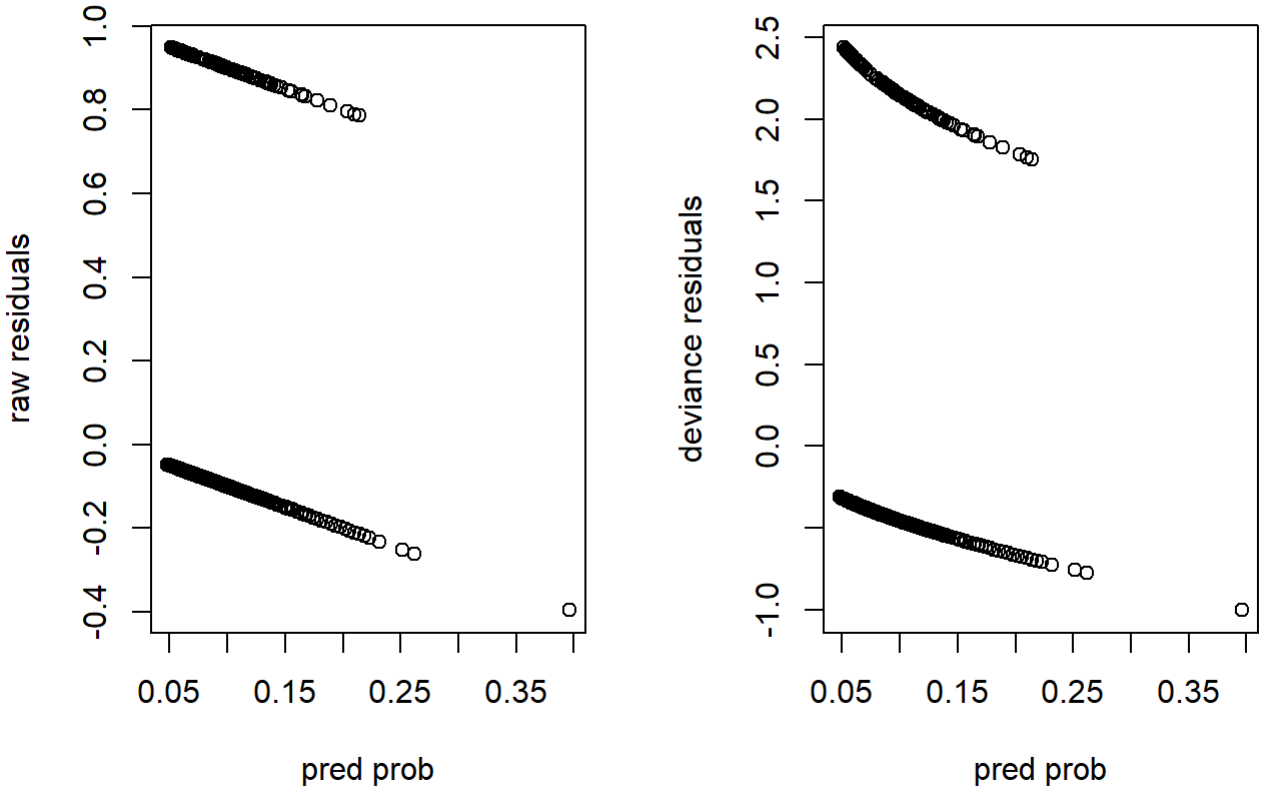
Let

$$\text{Deviance } = \sum_{i=1}^{n} r_i^2$$

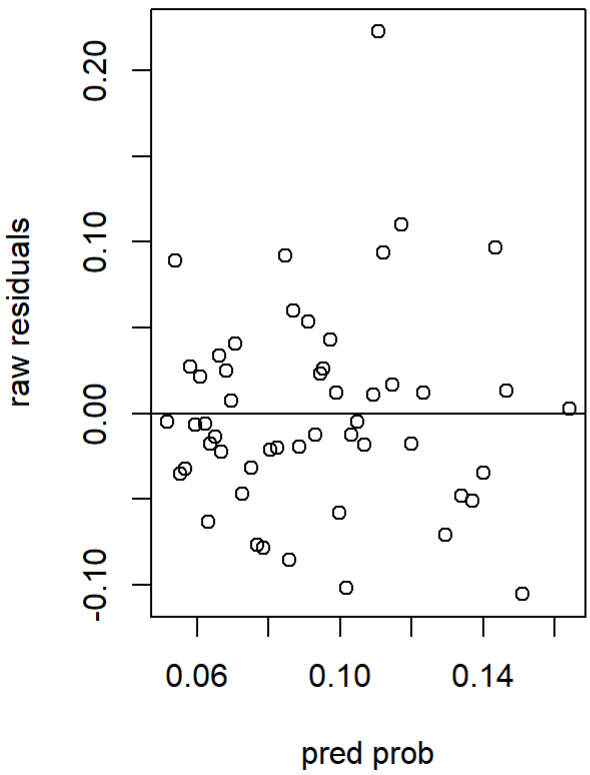
The deviance residuals for individual i, r_i:

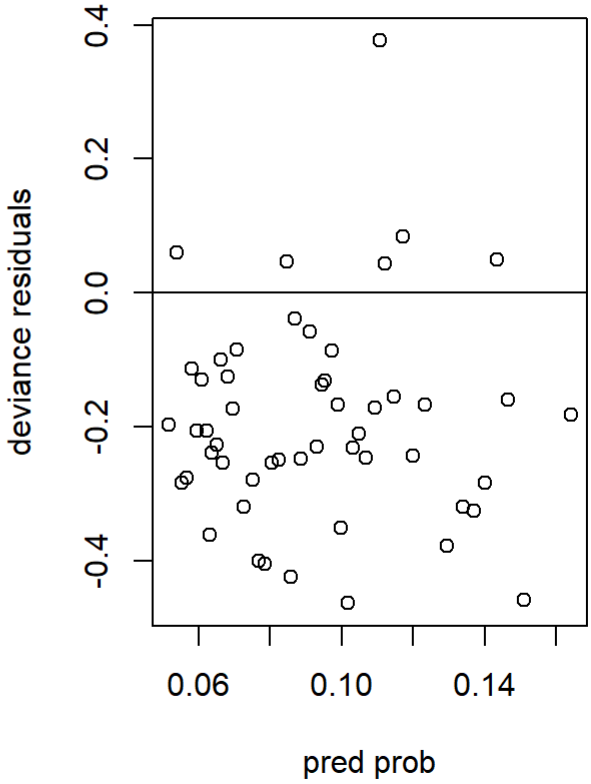
$$r_i = sign(y_i - \hat{p}_i)\sqrt{r_i}^2$$

```
# deviance residual
dres<-residuals(lmod)

# Calculating Bin Statistics
dres.bin<-tapply(dres, group, mean)

par(mfrow=c(1,2))
plot(predprob,rawres,xlab="pred prob",ylab="raw residuals")
plot(predprob,dres,xlab="pred prob",ylab="deviance residuals")
```




```
par(mfrow=c(1,2))
plot(predprob.bin,rawres.bin,xlab="pred prob",ylab="raw residuals")
abline(h=0)
plot(predprob.bin,dres.bin,xlab="pred prob",ylab="deviance residuals")
abline(h=0)
```





1.3 Bin residuals against the predictors

1.3.1 Height

```
table(wcgs$height)
```

```
##
##  60  61  62  63  64  65  66  67  68  69  70  71  72  73  74  75  76  77  78
##   5   3   5  20  41  60 181 232 412 412 556 419 409 195 122  53  23   3   3
```

```
rawres.bin.h<-tapply(rawres, wcgs$height, mean)
dres.bin.h<-tapply(dres, wcgs$height, mean)

dev.off()
```

```
## null device
##           1
```

```
plot(as.numeric(names(dres.bin.h)), dres.bin.h, xlab="height", ylab="deviance residuals")

wcgs$res <- dres
wcgs$phat <- predprob

wcgs[wcgs$height==77,c("cigs","height","y","phat","res")]
```

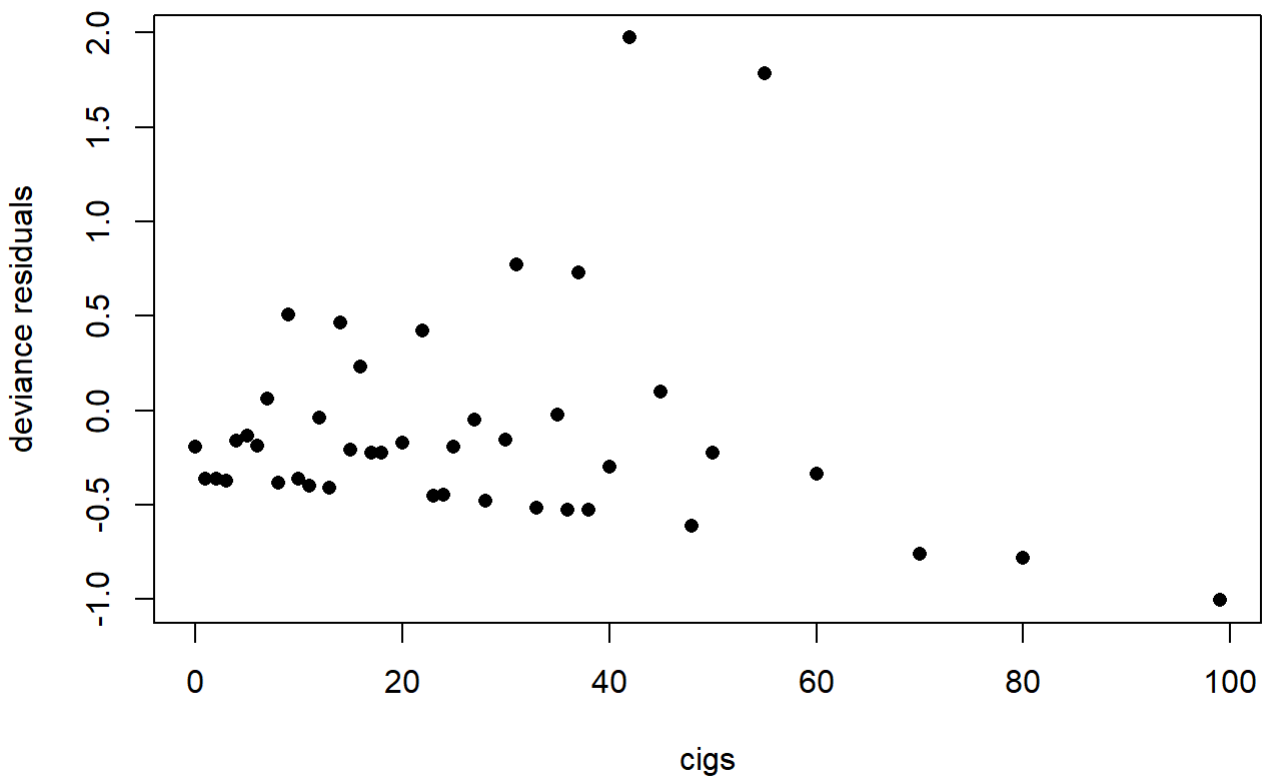
```
##      cigs height y       phat        res
## 1443    0     77 0 0.07171663 -0.3857933
## 2082    0     77 1 0.07171663  2.2956622
## 2634    5     77 0 0.07980662 -0.4078515
```

1.3.2 Number of cigarettes smoked per day

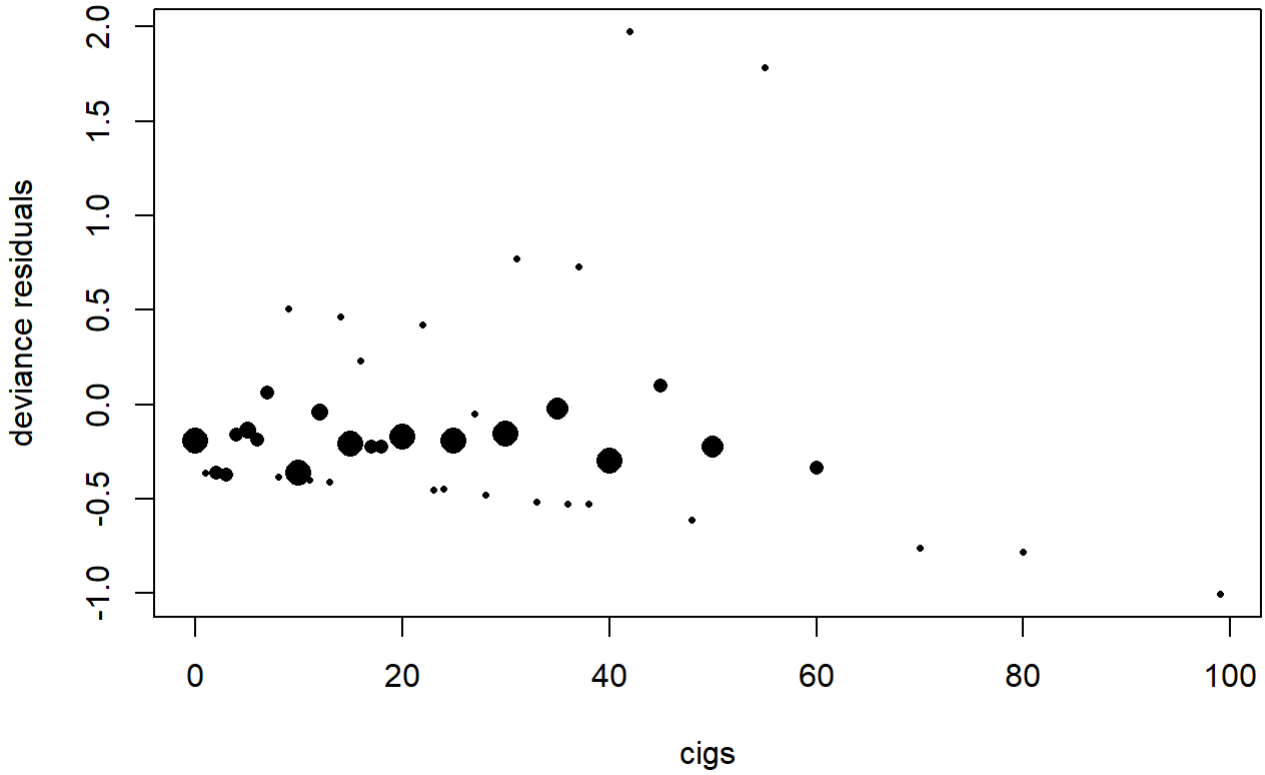
```
table(wcgs$cigs)
```

```
##
##    0    1    2    3    4    5    6    7    8    9   10   11   12   13   14   15
## 1652    7   15   14   13   23   14   12    4    3   83    3   22    4    3  102
##   16   17   18   20   22   23   24   25   27   28   30   31   33   35   36   37
##    4   13   13  453    3    5    1  105    6    2  283    2    4   36    2    2
##   38   40   42   45   48   50   55   60   70   80   99
##    1  182    1   11    1   32    1   14    1    1    1
```

```
dres.bin.c<-tapply(dres, wcgs$cigs, mean)

plot(as.numeric(names(dres.bin.c)), dres.bin.c, xlab="cigs", ylab="deviance residuals",pch=1
6)
```



```
count<-table(wcgs$cigs)
count.g<-cut(count,breaks=c(0,10,20,30,40,1700))
p.size<-ifelse(count>=40,1.8,0)
p.size[count<=10]<-.5
p.size[intersect(which(count>10), which(count<=20))]<-1
p.size[intersect(which(count>20), which(count<=30))]<-1.2
p.size[intersect(which(count>30), which(count<=40))]<-1.5

plot(as.numeric(names(dres.bin.c)), dres.bin.c, xlab="cigs", ylab="deviance residuals",pch=1
6,cex=p.size)
```



1.3.3 Detect cases which are unusual in the predictor space

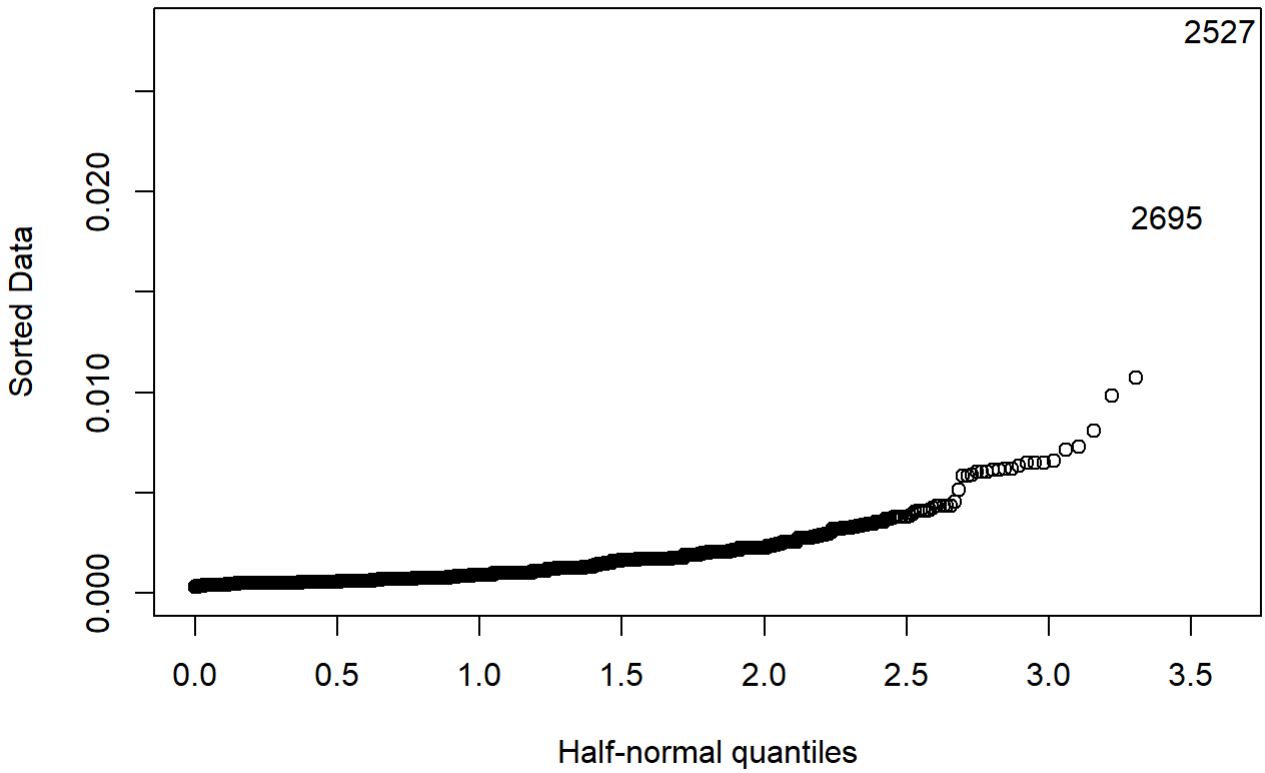
Examining the leverages (hat value). Leverage is a measure of how far an individual observation's predictor values are from the center of the data."

```
# half-normal plot
library(faraway)
```

```
##
## Attaching package: 'faraway'
```

```
## The following object is masked _by_ '.GlobalEnv':
##
##     wcgs
```

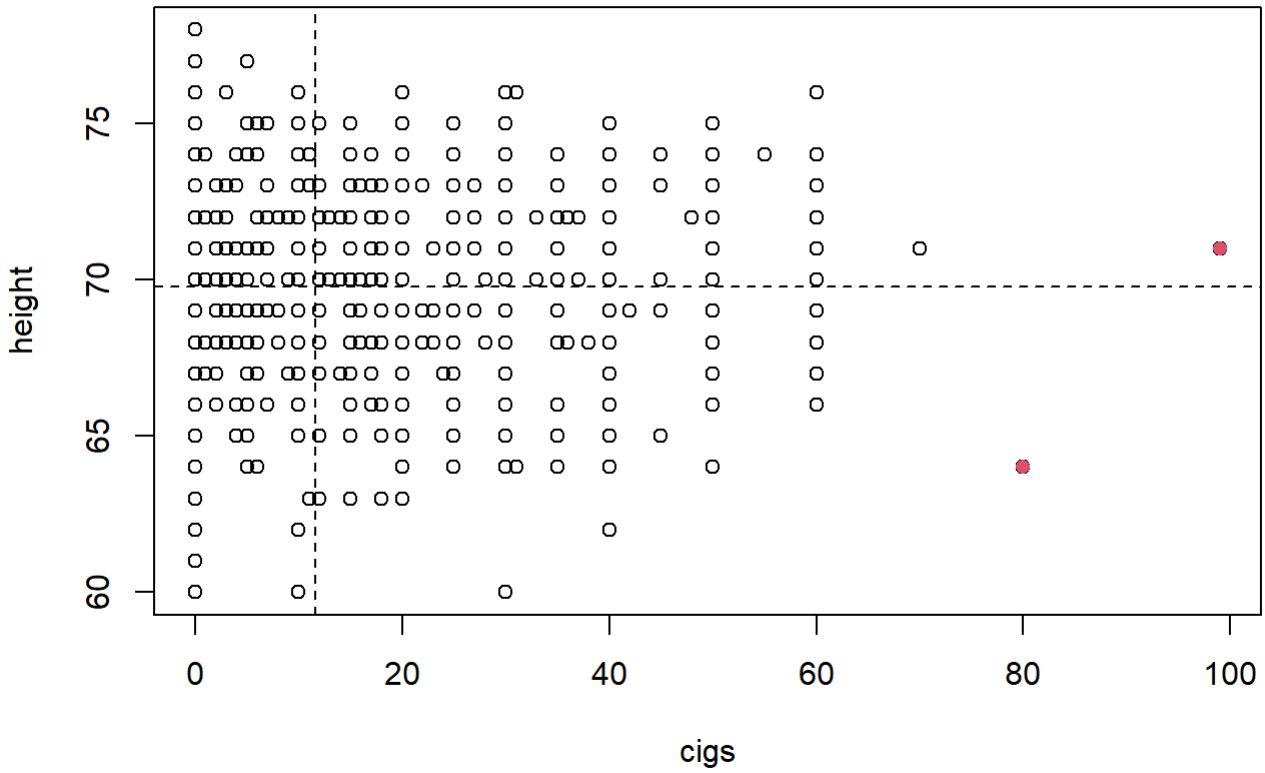
```
halfnorm(hatvalues(lmod))
```




```
wcgs[hatvalues(lmod) > 0.015,c("height","cigs","chd")]
```

```
##      height cigs chd
## 2527     71   99  no
## 2695     64   80  no
```

```
plot(wcgs$cigs, wcgs$height,xlab="cigs",ylab="height")

abline(v=mean(wcgs$cigs),lty=2)
abline(h=mean(wcgs$height),lty=2)
points(wcgs$cigs[hatvalues(lmod) > 0.015], wcgs$height[hatvalues(lmod) > 0.015],col=2,pch=16)
```


height
60
65
70
75
0
20
40
60
80
100
cigs